TERRACAST

Databricks Lakehouse Workflow for Weather-Driven Crop Yield Anomaly Detection

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Problem Statement

Crop yields vary across counties due to weather, irrigation, geography, and historical productivity. TERRACAST detects abnormal corn and soybean yield outcomes and explains whether they are linked to extreme weather.

Goal

Build an end-to-end Databricks workflow to detect, explain, and visualize county-level corn and soybean yield anomalies.

DATA INPUT

Yield Records	County × Year × Crop (NASS) Corn & Soybeans
Weather Variables	Temperature, rainfall, drought, flood, heat stress (PRISM, NOAA)
Geography	State/county FIPS Regional grouping
Management Signal	Irrigation status Historical productivity

Why Databricks?

Databricks Features Used

Lakeflow	Automates the data pipeline from raw data to curated feature tables.
MLflow	Tracks model experiments, evaluation metrics, and manages reproducible ML model versions.
Genie Space	Enables natural-language exploration of data and feature-driven patterns.
Databricks Apps	Deploys the interactive dashboard as a secure web application within Databricks.

End-to-End Workflow

1. **Data Ingestion** — Yield + weather + county data
2. **Lakeflow Pipeline** — Clean, join, transform
3. **Feature Engineering** — Weather stress + county baseline
4. **Model Training with MLflow** — Crop-region LightGBM + Random Forest ensemble
5. **Anomaly Detection & Explanation** — Residual z-score + SHAP
6. **Natural-Language Exploration with Genie Space** — Natural-language Q&A
7. **Interactive Dashboard with Databricks Apps** — Deploy an interactive dashboard for maps, predictions, SHAP explanations, and user exploration

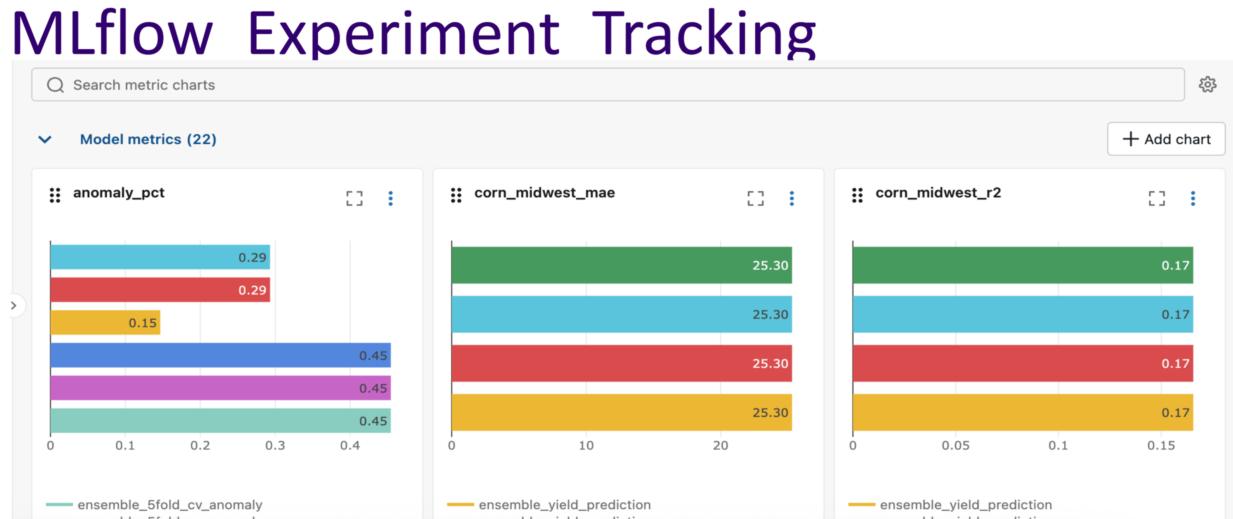
Model Performance

LightGBM + Random Forest ensemble

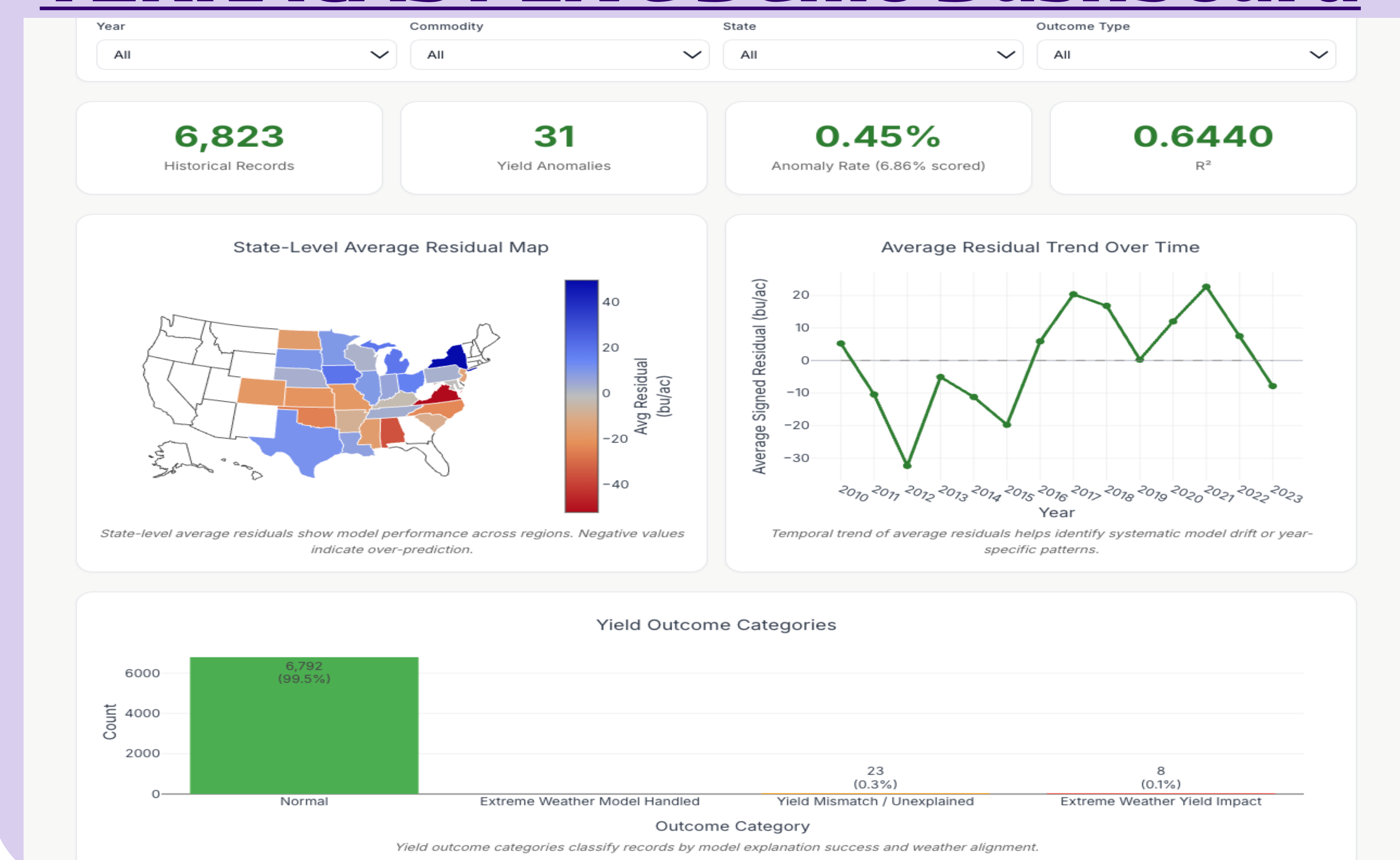
Metric	Result
R^2	0.6440
Yield Anomalies Detected	31
Anomaly Rate	0.45% overall 6.86% scored
Validation Strategy	County-based 5-fold CV

The ensemble achieved R^2 = 0.6440 and flagged 31 county-year-crop yield anomalies.

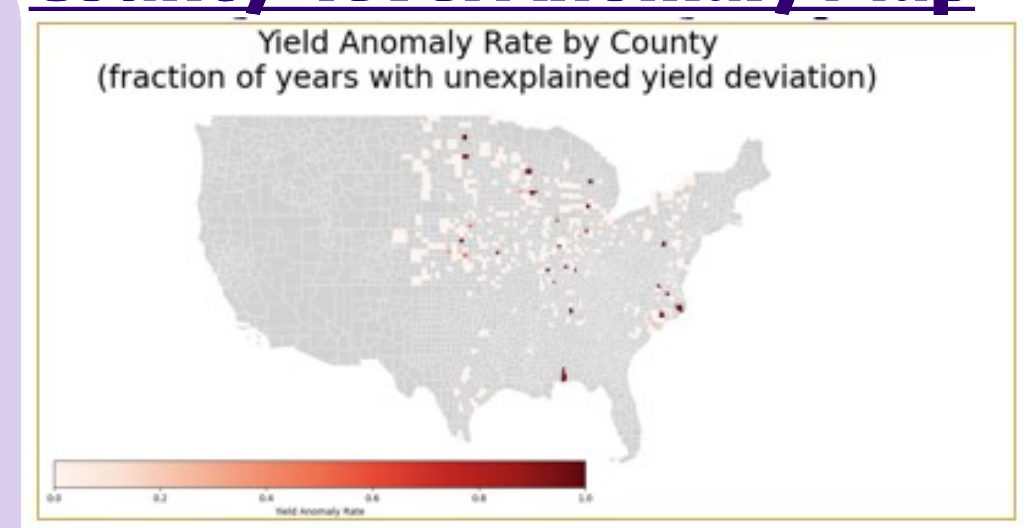
MLflow Experiment Tracking

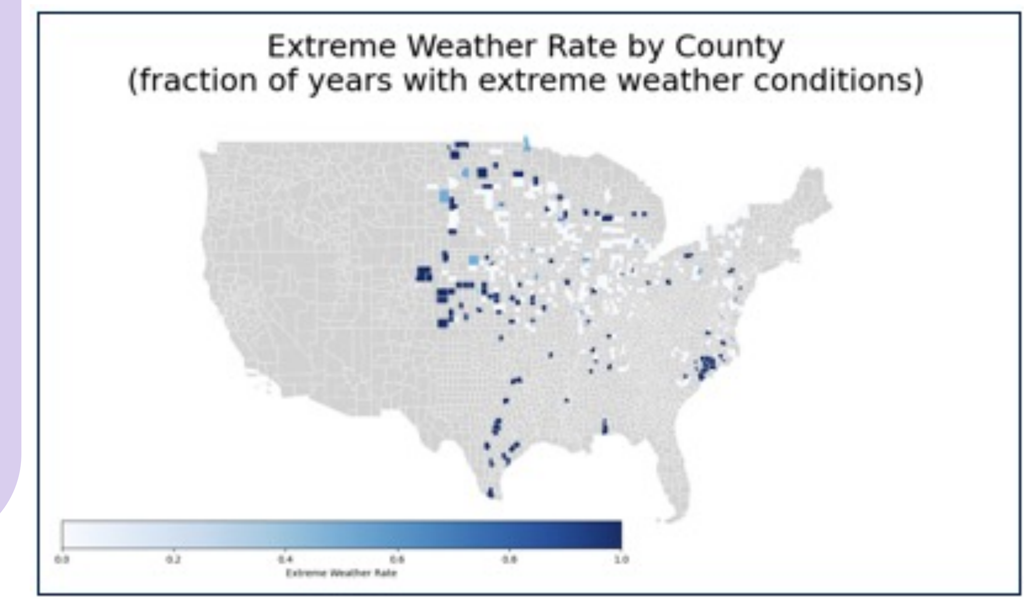


TERRACAST Live Demo Dashboard

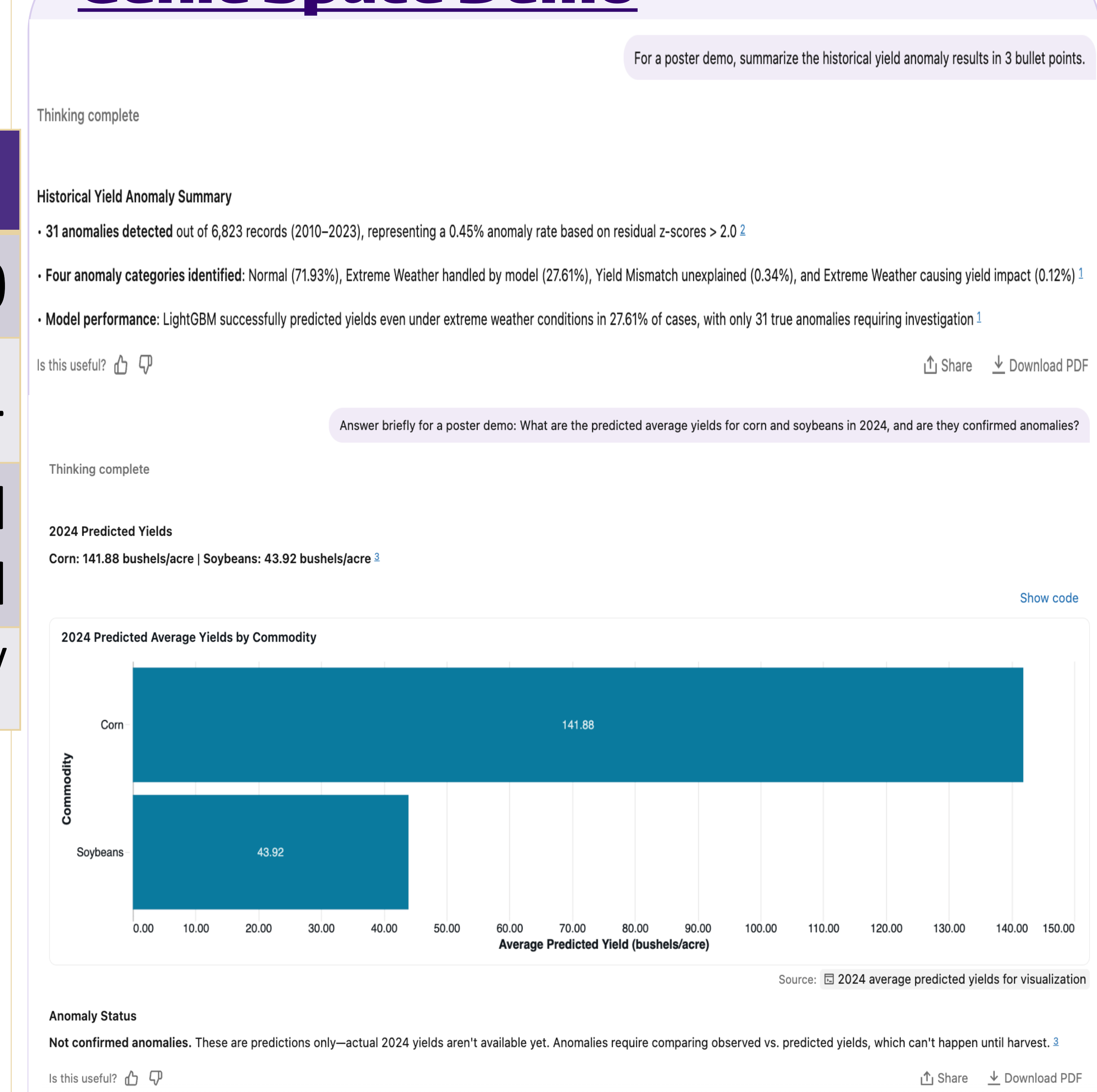


County-level Anomaly Map



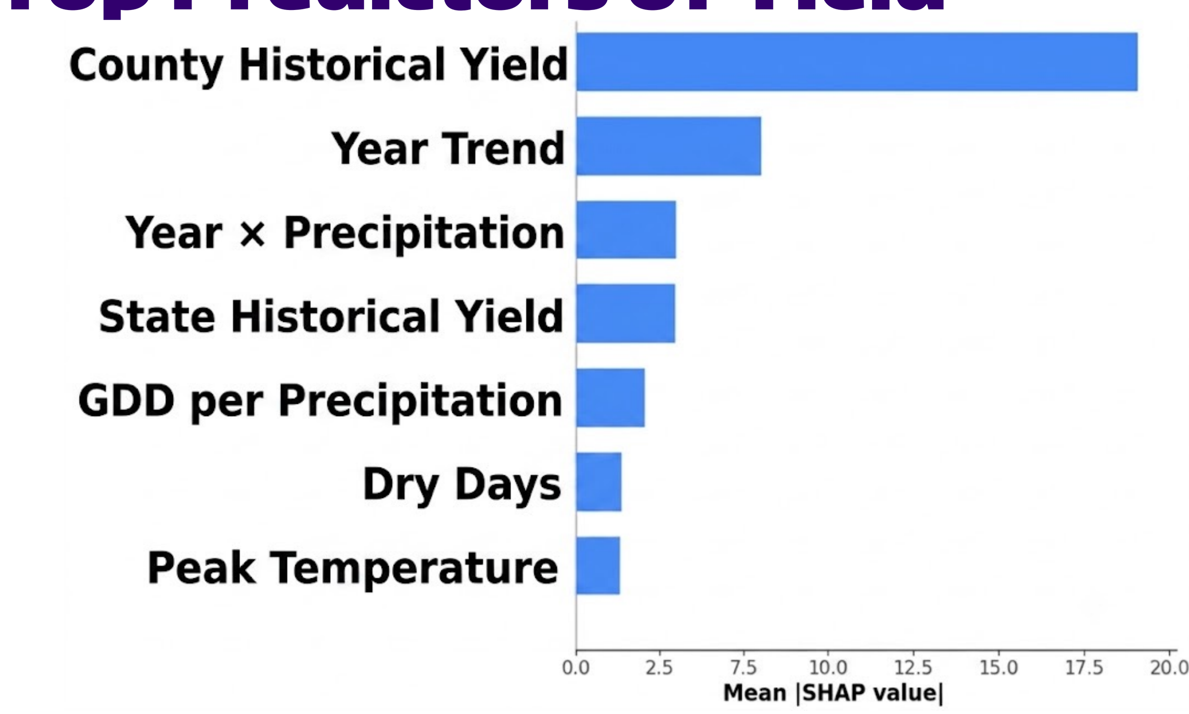


Genie Space Demo



TERRACAST detects yield outcomes that differ sharply from model-based predictions.

Top Predictors of Yield



Yield Outcome Categories